

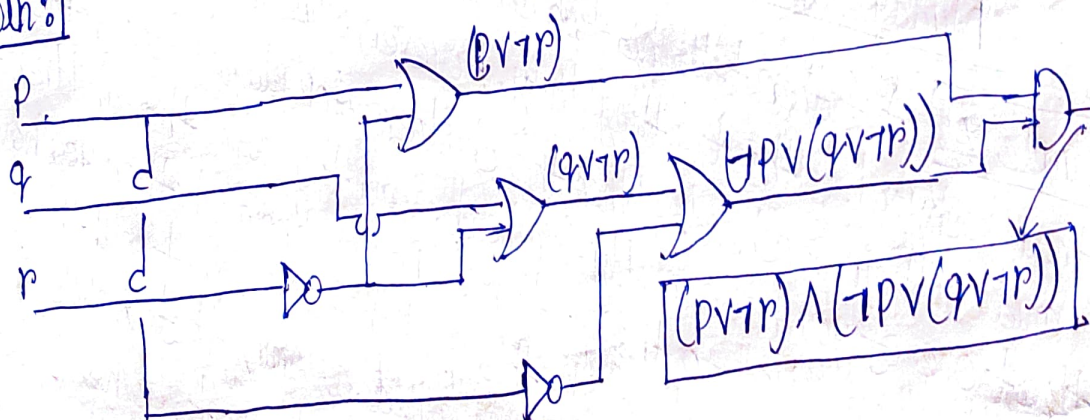
When asked for the second time, both will answer Yes. The boy will think since the girl answered No in the first time, it must be the case that I have a muddy forehead. Suppose, I didn't have a muddy forehead, she would automatically know I she has a muddy forehead based on the fact that father said "Atleast one of you has a muddy forehead" and he doesn't lie. She then would had answered Yes in the first time. The girl will think the same about the boy.

Answer: First time both simultaneously says ^{No} ~~Yes~~, Second time both simultaneously says Yes.

Logic Circuits

Example 11: Build a digital circuit that produces the output $(p \vee \neg r) \wedge (\neg p \vee (q \vee \neg r))$ when given input bits p, q and r.

Soln:



Exercises 1.2

In Exercises 1-6, translate the given statement into propositional logic using the propositions provided.

* You cannot edit a protected Wikipedia entry unless you are an administrator. Express your answer in terms of e: "You can edit a protected Wikipedia entry" and a: "You are an administrator."

Soln: $[e \rightarrow a] \wedge \neg \text{book}$ Me: $(\neg a \rightarrow \neg e) \wedge (a \rightarrow e)$